

G4 Differentiation rules A level only

Name: _____

Class: _____

Date: _____

Time: **76 minutes**

Marks: **63 marks**

Comments:

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Q1.

- (a) Differentiate $\ln x$ with respect to x . (1)
- (b) Given that $y = (x + 1) \ln x$, find $\frac{dy}{dx}$. (2)
- (c) Find an equation of the normal to the curve $y = (x + 1) \ln x$ at the point where $x = 1$. (4)
- (Total 7 marks)**

Q2.

- (a) A curve has equation $y = (x^2 - 3)e^x$.
- (i) Find $\frac{dy}{dx}$. (2)
- (ii) Find $\frac{d^2y}{dx^2}$. (2)
- (b) (i) Find the x -coordinate of each of the stationary points of the curve. (4)
- (ii) Using your answer to part (a)(ii), determine the nature of each of the stationary points. (2)
- (Total 10 marks)**

Q3.

- (a) Find $\frac{dy}{dx}$ when $y = \tan 3x$. (2)
- (b) Given that $y = \frac{3x+1}{2x+1}$, show that $\frac{dy}{dx} = \frac{1}{(2x+1)^2}$. (3)
- (Total 5 marks)**

Q4.

- (a) Find $\frac{dy}{dx}$ when $y = (3x - 1)^{10}$. (2)

- (b) Use the substitution $u = 2x + 1$ to find $\int x(2x + 1)^8 dx$, giving your answer in terms of x .

(4)

(Total 6 marks)

Q5.

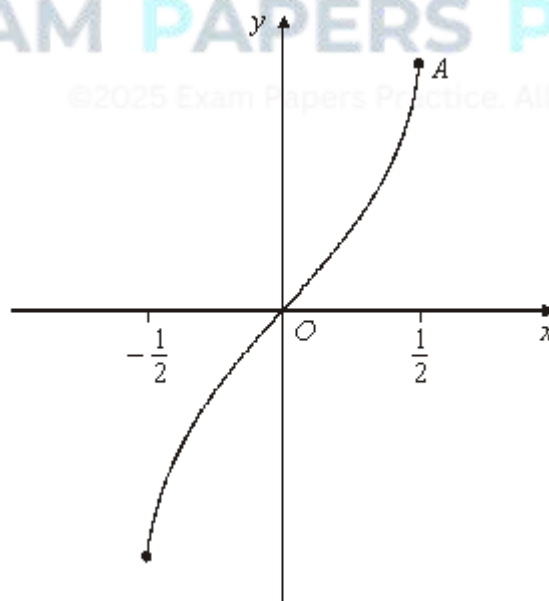
- (a) Given that $y = x \ln x$, find $\frac{dy}{dx}$. (2)

- (b) Hence, or otherwise, find $\int_1^{\ln} x dx$. (2)

- (c) Find the exact value of $\int_1^5 \frac{1}{x} dx$. (2)
- (Total 6 marks)

Q6.

The diagram shows the curve with equation $y = \sin^{-1} 2x$, where $-\frac{1}{2} \leq x \leq \frac{1}{2}$.



- (a) Find the y -coordinate of the point A , where $x = \frac{1}{2}$. (1)

- (b) (i) Given that $y = \sin^{-1} 2x$, show that $x = \frac{1}{2} \sin y$. (1)

- (ii) Given that $x = \frac{1}{2} \sin y$, find $\frac{dx}{dy}$ in terms of y . (1)

- (c) Using the answers to part (b) and a suitable trigonometrical identity, show that

$$\frac{dx}{dy} = \frac{2}{\sqrt{1-4x^2}}$$

(4)
(Total 7 marks)

Q7.

A curve is defined by the equation

$$y^2 - xy + 3x^2 - 5 = 0$$

- (a) Find the y -coordinates of the two points on the curve where $x = 1$. (3)

- (b) (i) Show that $\frac{dy}{dx} = \frac{y-6x}{2y-x}$. (6)

- (ii) Find the gradient of the curve at each of the points where $x = 1$. (2)

- (iii) Show that, at the two stationary points on the curve, $33x^2 - 5 = 0$. (3)
(Total 14 marks)

Q8.

- (a) Find $\frac{dy}{dx}$ when $y = x \sin 2x$. (3)

- (b) (i) Find $\frac{dy}{dx}$ when $y = (x^2 - 6)^4$.

(2)

(ii) Hence, or otherwise, find $\int x(x^2 - 6)^3 dx$.

(3)

(Total 8 marks)



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